

## CURIONEST

# Acids & Bases: Polyprotic Acids & Mixed Solutions

*Complete Lesson Pack · Stepwise Ionization ·  $K_{a1}$  &  $K_{a2}$  · Excess Moles*

### WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

*Made by a chemistry teacher — aligned to the standard US chemistry sequence.*

## Learning Objectives

1. Explain why polyprotic acids ionize stepwise and why each successive  $K_a$  is smaller
2. Perform equilibrium calculations for diprotic acids such as  $H_2CO_3$  and  $H_2S$
3. Apply the 'moles first, react to completion, divide by total volume' method to mixed-solution problems
4. Predict which species dominate at each stage of a polyprotic titration
5. Combine acid-base equilibria with stoichiometry in multi-step calculations

### Concept 1 — Polyprotic Acids and Stepwise Ionization

A polyprotic acid has more than one ionizable proton. It ionizes in steps, one proton at a time:  $H_2CO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$  ( $K_{a1} = 4.3 \times 10^{-7}$ ), then  $HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CO_3^{2-}(aq)$  ( $K_{a2} = 4.7 \times 10^{-11}$ ).

Each successive ionization constant is smaller than the previous one — typically by a factor of  $10^4$ – $10^5$  — because it is harder to remove a proton from a negatively charged ion.

Because  $K_{a1} \gg K_{a2}$ , the hydronium concentration in a solution of a diprotic acid is essentially determined by the first step alone. The second step contributes negligibly to  $[H_3O^+]$ .

The intermediate ion (e.g.,  $HCO_3^-$ ,  $HSO_4^-$ ) is amphiprotic: it can act as an acid ( $K_{a2}$ ) or a base ( $K_b = K_w/K_{a1}$ ).

#### ★ Worked Example

Calculate the concentration of  $H_3O^+$  and the pH of a 0.050 M solution of  $H_2S$ .  $K_{a1} = 1.0 \times 10^{-7}$ ,  $K_{a2} = 1.0 \times 10^{-19}$ .

**Solution:** Use only  $K_{a1}$ :  $H_2S + H_2O \rightleftharpoons H_3O^+ + HS^-$ .  $K_{a1} = x^2/(0.050 - x) \approx x^2/0.050 = 1.0 \times 10^{-7} \rightarrow x = \sqrt{(5.0 \times 10^{-9})} = 7.1 \times 10^{-5}$  M.  $pH = -\log(7.1 \times 10^{-5}) = 4.15$ . ( $K_{a2}$  is so small that the second step adds a negligible amount of  $H_3O^+$ .)

★ **Common mistake:** Writing both ionization steps in one equation — polyprotic acids ionize one proton at a time.

★ **Common mistake:** Adding the  $H_3O^+$  from the second step to the first step's result (negligible for typical problems).

### Concept 2 — Mixed-Solution Stoichiometry (the 'Moles First' Method)

When two solutions are mixed, the acid-base reaction is fast and goes to completion for strong acids/bases. Always: (1) convert to moles, (2) react to completion (find the limiting reactant and leftover), (3) divide by the TOTAL volume to get new concentrations, then (4) calculate pH from whatever remains.

If excess strong acid remains → pH from leftover  $[H_3O^+]$ . If excess strong base remains → pH from leftover  $[OH^-]$ . If exactly neutralized with a weak partner → solve the hydrolysis ( $K_b$  or  $K_a$ ) of the salt formed.

The total volume is the sum of both solution volumes — forgetting this is the most common error.

#### ★ Worked Example

50.0 mL of 0.100 M HCl is mixed with 50.0 mL of 0.060 M NaOH. Find the pH of the resulting solution.

**Solution:** Moles HCl = 0.00500; moles NaOH = 0.00300. HCl is in excess by 0.00200 mol. Total volume = 0.100 L.  $[H_3O^+] = 0.00200/0.100 = 0.020$  M.  $pH = -\log(0.020) = 1.70$ .

★ **Common mistake:** Using the original volumes instead of the total volume after mixing.

★ **Common mistake:** Neutralizing mole-for-mole without checking which reactant is in excess.

### Concept 3 — Polyprotic Titrations and Species Distribution

The titration curve of a diprotic acid (e.g.,  $H_3PO_4$ ,  $H_2CO_3$ ) shows two equivalence points and two buffer regions. Between equivalence points, the intermediate species ( $H_2PO_4^-$ ,  $HPO_4^{2-}$ ) buffers the solution.

At the first half-equivalence point,  $[H_3PO_4] = [H_2PO_4^-]$  and  $pH = pK_{a1}$ . At the second half-equivalence point,  $[H_2PO_4^-] = [HPO_4^{2-}]$  and  $pH = pK_{a2}$ .

Which species dominates depends on pH:  $H_3PO_4$  dominates below  $pK_{a1}$ ,  $H_2PO_4^-$  dominates between  $pK_{a1}$  and  $pK_{a2}$ , and  $HPO_4^{2-}$  dominates above  $pK_{a2}$ .

#### ★ Worked Example

For phosphoric acid ( $pK_{a1} = 2.15$ ,  $pK_{a2} = 7.20$ ,  $pK_{a3} = 12.35$ ), which species dominates at (a) pH 1.0 (b) pH 5.0 (c) pH 9.0?

**Solution:** (a)  $H_3PO_4$  (below  $pK_{a1}$ ) (b)  $H_2PO_4^-$  (between  $pK_{a1}$  and  $pK_{a2}$ ) (c)  $HPO_4^{2-}$  (between  $pK_{a2}$  and  $pK_{a3}$ ).

★ **Common mistake:** Expecting a single equivalence point for a polyprotic acid.

★ **Common mistake:** Choosing an indicator that changes color at the wrong equivalence point.

**Part A — Polyprotic Acid Calculations (Questions 1–5)**

1. Explain why  $K_{a2}$  is always smaller than  $K_{a1}$  for a polyprotic acid such as  $H_2CO_3$ .
2. Write the two stepwise ionization equations for  $H_2CO_3$ , labeling  $K_{a1}$  and  $K_{a2}$ .
3. Which species are present in appreciable concentration in a solution of  $H_2CO_3$ ? Explain why  $CO_3^{2-}$  is present in only trace amounts.
4. Calculate the  $H_3O^+$  concentration and pH of a 0.010 M solution of  $H_2S$ .  $K_{a1} = 1.0 \times 10^{-7}$ ,  $K_{a2} = 1.0 \times 10^{-19}$ .
5. For phosphoric acid ( $pK_{a1} = 2.15$ ,  $pK_{a2} = 7.20$ ,  $pK_{a3} = 12.35$ ), identify the dominant species at pH 1.0, pH 5.0, pH 9.0, and pH 13.0.

**Part B — Mixed-Solution Stoichiometry (Questions 6–10)**

6. 25.0 mL of 0.200 M HCl is mixed with 75.0 mL of 0.050 M NaOH. Calculate the pH of the mixture.
7. 40.0 mL of 0.150 M KOH is mixed with 60.0 mL of 0.080 M HCl. Calculate the pH of the mixture.
8. 30.0 mL of 0.100 M  $H_2SO_4$  (diprotic, both protons strong) is mixed with 70.0 mL of 0.050 M NaOH. Calculate the pH.
9. 50.0 mL of 0.100 M  $CH_3COOH$  is mixed with 25.0 mL of 0.100 M NaOH.  $K_a = 1.8 \times 10^{-5}$ . Is the mixture a buffer? Calculate its pH.
10. 100.0 mL of 0.050 M HCN ( $K_a = 6.2 \times 10^{-10}$ ) is mixed with 50.0 mL of 0.100 M NaOH. Calculate the pH of the resulting solution.

**Part C — Weak-Acid Equilibrium (ICE Table Review) (Questions 11–13)**

11. Propionic acid,  $C_2H_5CO_2H$  ( $K_a = 1.34 \times 10^{-5}$ ), is used to make the food preservative calcium propionate. What is the pH of a 0.698 M solution of  $C_2H_5CO_2H$ ?
12. White vinegar is about 5.0% acetic acid by mass (density 1.007 g/mL). Estimate its pH. ( $CH_3COOH$ :  $K_a = 1.8 \times 10^{-5}$ , molar mass 60.05 g/mol.)

13. The pH of a 0.10 M solution of caffeine is 11.70. Determine  $K_b$  for caffeine.

## Key Terms

| Term                                | Definition                                                                                             |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>polyprotic acid</b>              | An acid with more than one ionizable proton (e.g., $\text{H}_2\text{CO}_3$ , $\text{H}_3\text{PO}_4$ ) |
| <b>stepwise ionization</b>          | Ionization of a polyprotic acid one proton at a time, with separate $K_a$ values                       |
| <b>amphiprotic intermediate</b>     | The ion between ionization steps (e.g., $\text{HCO}_3^-$ ) that can act as acid or base                |
| <b>mixed-solution stoichiometry</b> | The moles-first method for acid-base reactions of mixed solutions                                      |
| <b>limiting reactant</b>            | The reactant that is used up first, determining how much product forms                                 |
| <b>species distribution</b>         | Which protonated form of a polyprotic acid dominates at a given pH                                     |

## Answer Key

Worked solutions for every question — see answers embedded in Parts A–C above. The full key with step-by-step work is included in the teacher edition.

### Part A — Polyprotic Acid Calculations (Questions 1–5)

1. Removing a proton from a negatively charged ion (e.g.,  $\text{HCO}_3^- \rightarrow \text{CO}_3^{2-}$ ) requires more energy than removing one from a neutral molecule. Electrostatic attraction makes the remaining proton harder to remove.
2.  $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$  ( $K_{a1}$ );  $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$  ( $K_{a2}$ )
3.  $\text{H}_2\text{CO}_3$ ,  $\text{HCO}_3^-$ ,  $\text{H}_3\text{O}^+$ , and  $\text{OH}^-$ .  $\text{CO}_3^{2-}$  is present only in trace amounts because  $K_{a2}$  ( $4.7 \times 10^{-11}$ ) is tiny — very little  $\text{HCO}_3^-$  ionizes further.
4.  $x^2/0.010 = 1.0 \times 10^{-7} \rightarrow x = 3.2 \times 10^{-5} \text{ M}$ ;  $\text{pH} = 4.50$ . (Second step negligible.)
5.  $\text{pH } 1.0$ :  $\text{H}_3\text{PO}_4$  ·  $\text{pH } 5.0$ :  $\text{H}_2\text{PO}_4^-$  ·  $\text{pH } 9.0$ :  $\text{HPO}_4^{2-}$  ·  $\text{pH } 13.0$ :  $\text{PO}_4^{3-}$

### Part B — Mixed-Solution Stoichiometry (Questions 6–10)

6. Moles  $\text{HCl} = 0.00500$ ; moles  $\text{NaOH} = 0.00375$ . Excess  $\text{HCl} = 0.00125 \text{ mol}$  in  $0.100 \text{ L} \rightarrow [\text{H}_3\text{O}^+] = 0.0125 \text{ M}$ ;  $\text{pH} = 1.90$ .
7. Moles  $\text{KOH} = 0.00600$ ; moles  $\text{HCl} = 0.00480$ . Excess  $\text{KOH} = 0.00120 \text{ mol}$  in  $0.100 \text{ L} \rightarrow [\text{OH}^-] = 0.0120 \text{ M}$ ;  $\text{pOH} = 1.92$ ;  $\text{pH} = 12.08$ .
8.  $\text{H}^+$  from  $\text{H}_2\text{SO}_4 = 0.0300 \text{ L} \times 0.100 \text{ M} \times 2 = 0.00600 \text{ mol}$ .  $\text{OH}^- = 0.0700 \times 0.050 = 0.00350 \text{ mol}$ . Excess  $\text{H}^+ = 0.00250 \text{ mol}$  in  $0.100 \text{ L} \rightarrow [\text{H}_3\text{O}^+] = 0.0250 \text{ M}$ ;  $\text{pH} = 1.60$ .
9. Yes — a buffer forms. Moles acid =  $0.00500$ , moles  $\text{NaOH} = 0.00250 \rightarrow [\text{CH}_3\text{COOH}] = 0.00250/0.0750 = 0.0333 \text{ M}$ ,  $[\text{CH}_3\text{COO}^-] = 0.00250/0.0750 = 0.0333 \text{ M}$ .  $\text{pH} = \text{pK}_a + \log(1) = 4.74$ .
10. Exactly neutralized: moles  $\text{HCN} = 0.00500$ , moles  $\text{NaOH} = 0.00500 \rightarrow \text{all HCN} \rightarrow \text{CN}^-$ .  $[\text{CN}^-] = 0.00500/0.150 = 0.0333 \text{ M}$ .  $K_b = K_w/K_a = 1.6 \times 10^{-5}$ .  $[\text{OH}^-] = \sqrt{(1.6 \times 10^{-5} \times 0.0333)} = 7.3 \times 10^{-4} \text{ M}$ ;  $\text{pOH} = 3.14$ ;  $\text{pH} = 10.86$ .

### Part C — Weak-Acid Equilibrium (ICE Table Review) (Questions 11–13)

11.  $x^2/0.698 = 1.34 \times 10^{-5} \rightarrow x = \sqrt{(9.35 \times 10^{-6})} = 3.06 \times 10^{-3} \text{ M}$ .  $\text{pH} = -\log(3.06 \times 10^{-3}) = 2.51$ .
12.  $5.0\% \text{ by mass} \approx 50 \text{ g/L} \rightarrow [\text{CH}_3\text{COOH}] = 50/60.05 = 0.83 \text{ M}$ .  $x = \sqrt{(1.8 \times 10^{-5} \times 0.83)} = 3.9 \times 10^{-3} \text{ M}$ ;  $\text{pH} = 2.41$ .
13.  $\text{pOH} = 14.00 - 11.70 = 2.30 \rightarrow [\text{OH}^-] = 5.0 \times 10^{-3} \text{ M}$ .  $K_b = (5.0 \times 10^{-3})^2/(0.10 - 5.0 \times 10^{-3}) = 2.6 \times 10^{-4}$ .